Q-Module-Bot: A Generative AI-Based Question and Answer Bot for Module Teaching Support

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Abstract—Contributions: **In this article, a generative artificial intelligence (AI)-based Q&A system has been developed by integrating information retrieval and natural language processing techniques, using course materials as a knowledge base and facilitating real-time student interaction through a chat interface.**

Background: **The rise of advanced AI exemplified by ChatGPT developed by OpenAI, has sparked interest in its application within higher education. AI has the potential to reshape education delivery through chatbots and related tools, improving remote learning and mitigating challenges, such as student isolation and educator administrative burdens. Yet, ChatGPT's practical applications in education remain uncertain, potentially due to its novel and enigmatic nature. Additionally, current e-learning chatbot systems often suffer from development complexity and a lack of input from key stakeholders, leading to developer-focused solutions rather than user-centered ones.**

Intended Outcomes: **In this manuscript, we introduce a practical implementation of AI in education by creating a system called *Q-Module-Bot* that is accessible for both technical and nontechnical educators to harness e-learning benefits and demystify generative pretraining transformer (GPT).**

Application Design: **The proposed Q-Module-Bot system has utilized pretrained large language models (LLMs) to build a Q&A system that helps students with their queries and supports education delivery using content extracted from a virtual learning environment (VLE).**

Findings: **The prototype and system evaluation confirm the effectiveness of a scalable cross-departmental tool featuring source attribution and real-time responses. While successful in encouraging wider acceptance of GPT use cases in higher education, refinements are needed for full integration into the VLE and expansion to other modules/courses.**

Index Terms**—Artificial intelligence (AI), chatbots, ChatGPT, e-learning, generative AI, information retrieval (IR), virtual learning environment (VLE).**

I. Introduction

THE SURGE in online learning, accelerated by the COVID-19 pandemic, has driven a need for broader adoption of digitally enabled higher-education services [1], [27], [28]. Virtual learning environments (VLEs) emerged as valuable platforms for facilitating interaction and engagement in education delivery. The increasing prevalence of digital education services has brought to the forefront the critical need to equip educators, students, and academic support teams with the appropriate tools to operate in fully remote or hybrid settings seamlessly. As such, it has become of utmost importance to ensure that all stakeholders (students, teaching and administrative staff) have access to the requisite resources to enable them to participate actively in the digital learning environment [2], [29], [30]. Higher-education services commonly offer technologies that support online student communication, such as video technology for managing remote lectures, emails, and forums for queries. However, limitations, such as minimal online student participation in video meetings and forums, as well as delayed academic responses to student queries from online messages and emails, are common, as well as the capacity for training and support required to effectively utilize these systems, which pose significant challenges for many institutions [7], [24].

Artificial intelligence (AI) technology presents an opportunity to alleviate some of these challenges, potentially reducing administrative burden and mitigating associated stress related to those responsible for delivering education services [18]. AI and chatbots have been introduced in the education sector as one solution to support teaching services and enhance student experiences online, and existing solutions span from support bots to virtual teaching assistants [32]. Nevertheless, neglecting stakeholder perspectives during the developmental phase, which encompasses content delivery and student support roles, including those who would eventually implement and manage these tools, could result in less efficient systems that fail to align with user needs [33]. Hence, it remains imperative to incorporate the valuable insights of academic educators and teaching services stakeholders throughout the requirements-gathering process [14], [22]. Furthermore, current solutions disregard the critical aspect of tailoring the system to module-specific requirements that can be scaled across modules/courses, and equally essential is the provision of user-friendly interfaces designed to accommodate educators who lack technical expertise yet intend to use the system to deliver their own content effectively. Integrating novel technology in education can face obstacles due to uncertainty surrounding attitudes and knowledge of its real-world applications within universities [34]. In some institutions, departments choose to steer clear of technology implementation into teaching [13], with those who oppose digital adoption arguing they see no notable impact on the learning outcomes when technology is incorporated into the educational

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process [21]. The integration of educational technology in higher education for the benefit of both educators and students does not always lead to seamless integration into instruction, and recommendations for integration include the establishment of clear goals, which include understanding stakeholders' attitudes toward technology, integrating strategies to reshape negative perceptions and working on what good adoption looks like [13], [21]. Furthermore, there are arguments for those delivering education to move focus from whether to use technology for instruction to determine how to successfully integrate and spread technology for educational enhancement while comprehending the context of technology and education and meeting the evolving needs of students [21]. These complexities often lead to a decelerated adoption of innovative approaches, further compounded by the imperative need to address ethical concerns. However, to facilitate the adoption of a solution, it is imperative that stakeholders are aligned with the intended outcomes, the solution is put through testing, and its use is both convenient and user-friendly [17].

A. Main Contributions

The work in this article presents "Q-Module-Bot," a tool that is specifically designed for educational delivery. The tool utilizes a knowledge base that is sourced from course material (namely, the Electronic Engineering and Computer Science Capstone Project module) that has been scraped from Queen Mary University of London (QMUL)'s VLE known as "QMPlus." The primary objective of this tool is to address pedagogical challenges encountered in academic settings. These challenges include limited student engagement with module content and inadequate feedback loops between students and educators. As a result, academic responses to student queries from online messages and emails are often delayed. Q-Module-Bot has the potential to enhance student engagement and accessibility while also alleviating the burden on educators. This system provides actionable insights for educators, empowering them with the necessary tools to improve their instructional practices. This includes the ability to replicate the AI chatbot, thereby enabling customization and adaptation to individual teaching environments. The system is the first of its kind to utilize a generative pretraining transformer (GPT) specifically tailored for taught university modules and is supported by evaluation data from staff and students at QMUL. This represents an innovative approach to enhancing educational quality in the Electronic Engineering and Computer Science curriculum by leveraging pretrained large language models (LLMs) for a Question and Answer (Q&A) system. This system employs information retrieval (IR) and natural language processing (NLP) methods to efficiently deliver answers, marking a novel implementation of GPT-3 turbo for educational purposes.

The remainder of this article is structured as follows: Section II explores the context of recent advancements in AI and chatbot applications in education, examines existing efforts, assesses available technology frameworks, and presents stakeholder user study results. Section III outlines primary and secondary research-derived high-level requirements, followed by a strengths, weaknesses, opportunities, threats (SWOTs) analysis that informs the system's technical design and implementation, including the data pipeline. Section IV presents a user system evaluation, followed by Sections V and VI that conclude this article and suggest avenues for future development.

II. Background

This section examines the background of recent developments in AI and chatbot applications in education, reviews current initiatives, evaluates accessible technological frameworks, and presents stakeholder user research findings.

A. AI Chatbots and ChatGPT

The development of AI and chatbot technology has evolved significantly since its inception in the 1950s, with a huge estimated market size of 1.23 billion USD expected by 2025 [8]. Early breakthroughs with chatbots, such as the creation of the ELIZA Program in the 1960s, marked the start of an exploration of AI's potential to emulate and participate in human conversation, leading to the emergence of sophisticated chatbots capable of understanding natural language and responding to complex queries [9]. In the past year, OpenAI's ChatGPT has revolutionized the space and catapulted the technology to a global platform. ChatGPT utilizes GPT architecture, leveraging NLP and machine learning (ML) techniques to facilitate seamless and intelligent conversational interactions with users. Unlike previous AI, ChatGPT is a generative AI that can produce new content in real-time conversations. It has the ability to maintain a persona, allowing for more realistic and engaging dialogues as a result of training on vast conversational datasets [12]. The release of ChatGPT has gained significant attention since the release of GPT-3 and Davinci models in late 2022. ChatGPT managed to amass 100 million monthly active users in just two months since its launch, emphasizing the immediate impact and future potential of the technology [35]. This unprecedented growth solidifies its position as the fastest-growing online user application ever recorded with an average of around 13 million unique visitors engage with ChatGPT daily [20]. With its remarkable ability to comprehend intricate requests, ChatGPT is changing the way we engage with technology across the world, introducing possibilities for enhanced user experiences as industries explore diverse use cases [9].

B. Chatbots in Higher-Education Delivery

Education is one of many sectors that has adopted AI chatbots within institutions to support business operations and efficiency and ultimately improve student learning outcomes [6], [31]. The popularity of platforms and tools which support online learning is not new, with prepandemic levels of students enrolled in online or hybrid university courses at 6.7 million in the U.S., and 32% of all university student enrolments to a fully online course recorded as early as 2012 [15]. It is noteworthy that the recent emergence of disruptive language models, coupled with their user-friendly chat interfaces, has had a profound impact on the academic

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landscape. However, there are concerns regarding their application, particularly in relation to online examinations and coursework plagiarism, with arguments that ChatGPT could be used to cheat on essay assignments and online exams [9]. Plagiarism is a growing issue in higher education due to students' growing dependence on online sources for research and information, enabled by easy accessibility, resulting in a surge in behaviors lacking citation and subsequent serious consequences, including academic failure [9]. The technology has attracted significant media attention, as exemplified by a recent media article outlining the "educators battle" against plagiarism, describing ChatGPT as "the greatest cheating tool ever invented" [23]. A survey conducted earlier this year by Study.com, a VLE platform, with over 100 educators and 1000 students of college and higher-level education, reports that 89% of students have used ChatGPT to help with a homework assignment [24]. To offer guidance to educators on the use of technology, OpenAI [19] has issued recommendations for integrating technology into education. These suggestions encompass drafting lesson plans, designing exercises, and providing feedback on writing and coding. OpenAI also affirms it is essential to acknowledge that there are education-related risks, and there is an emphasis on introduction to the classroom under the supervision of educators. The rapid advancement of AI technology has brought about a shift in the traditional notions of attribution and originality. Consequently, it has become imperative for education providers to facilitate the integration of OpenAI's technology into university classrooms. Such integration should be done in recognition of the fact that students are already utilizing this technology. It is essential to tap into the full potential of OpenAI's technology while ensuring its implementation aligns with the educational objectives and requirements of the institutions [9].

C. AI Chatbots and ChatGPT

Prior studies on e-learning technology adoption emphasize the significance of forming an online community to enhance student satisfaction with online learning, as community-building fosters an environment akin to physical classrooms, promoting inclusivity and social support among peers and lecturers, facilitating information exchange and leading to positive student outcomes [15]. In higher education, learner-instructor interaction refers to the exchange between lecturers and students and is relevant to face-to-face and online teaching approaches. In the context of learner-instructor interaction, the instructor's primary responsibilities include igniting student interest in the course content, motivating learners, and facilitating the learning process [6]. Student satisfaction levels in an online taught university reveal that satisfaction is closely tied to the fulfilment of a student's interpersonal communication needs, as they highly value communication and individual, personal responses as part of their interactions [5]. This is made complex in higher-education settings, where class sizes can play a crucial role in shaping the dynamics of academic instruction and student engagement. Smaller class sizes tend to promote improved and more intimate interactions between students and educators, fostering stronger rapport. However, the reality in many universities is that class sizes are often quite large, with a ratio as high as one student to tens or hundreds for lecturers, depending on the course and institution. Consequently, creating meaningful personal interactions becomes inherently more challenging [4]. Despite the benefits of student forums and discussion boards, which are features of many VLEs and for remote courses can facilitate building relationships and communication channels between instructor and student [7], they are often under-utilized due to challenges native to online public forums, such as students feeling uncomfortable about asking questions in public view of other students as well as delayed response times [24]. Educational chatbots are gaining popularity as a cost-effective solution for delivering personalized learning experiences when it comes to higher education [10] and is arguably one of the most exciting and engaging tools to promote these experiences through VLE platforms [11]. Students are highly adept at using various platforms and receiving responses in real-time, such as Meta-owned WhatsApp, Facebook, and Instagram for communication. Chatbots represent a valuable opportunity to provide university-related information to students instantaneously. Such information may include but is not limited to, assessment criteria, due dates, and other relevant resources. While various means of interaction exist, such as email, student forums, and in-person communication, these methods lack the capability to offer personalized communication that is readily accessible at all times [4]. Personalization can drive outcomes that facilitate relationships and trust, improve student engagement, and support improvements to student satisfaction with e-learning [7].

D. Chatbots and Supporting Academics

Chatbot systems have the potential to not only enhance student engagement and provide valuable support to students but also significantly reduce the administrative workload of academics. By leveraging the capabilities of chatbots, these systems can effectively automate routine administrative tasks, such as answering frequently asked questions (FAQs), providing course materials, and managing student inquiries [4], [11]. This automation liberates academics from time-consuming administrative responsibilities, granting them more time and energy to dedicate to crucial aspects of their profession, such as curriculum development and research. Large class sizes in university courses can impose significant administrative challenges on lecturers, resulting in limited time to address student emails and responses to student forums, which may get lost in overflowing inboxes [4].

E. Chatbot Types, Capabilities, and Applications

To shape the approach for this implementation, it is important to consider the various types of technologies available to develop chatbots and their distinct capabilities, which include both rule-based and ML-based chatbots [23]. Rule-based chatbots rely on predefined developer-set rules, while ML-based chatbots are more advanced, leveraging AI to understand both commands and natural language. Rule-based chatbots have limitations and may fail to comprehend queries that do

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not match their predefined data. On the other hand, ML-based chatbots continuously improve by learning from past interactions with users [16]. In some cases, a combination of both techniques is incorporated depending on the requirements of the system. In terms of applications of these types of systems, Q&A bots are tailored to deliver precise answers from a knowledge base to specific queries suitable for customer support and FAQs. In contrast, chatbots engage in natural conversations, handling diverse queries, such as open-ended questions and casual conversation, making them versatile for applications like customer service, virtual assistants, and entertainment. However, while chatbots offer dynamic interactions, they may not match the accuracy of Q&A bots in addressing specific questions due to their focus on sustaining natural conversations. For "Q-Module-Bot," the focus is to respond to specific user queries, hence a Q&A bot seems to best fit this use case.

F. SWOT Analysis of Existing Applications

A qualitative analysis of chatbots in higher-education e-learning is analyzed through a SWOTs framework, detailed in Appendix A in the supplementary material. Adaptability and learning capabilities, as demonstrated in systems developed by Sinha et al. [16] and Yang and Stansfield [18], can lead to improved performance of the bot over time. However, reliance on predefined responses in rules-based systems, as observed in Ranoliya et al. [3], may limit handling complex queries. Overcoming technical complexities highlighted in Yang and Stansfield [18] and Sinha et al. [16] is crucial for user-friendly implementation. The precision and accuracy of responses, reported in systems developed by Sinha et al. [16], significantly impact user satisfaction, suggesting that students are keen to know the responses provided are accurate. The emergence of competing solutions, understood by Yang and Stansfield [18] and Ranoliya et al. [3], poses opportunities and threats, driving developers to explore novel technologies. User-centered design and stakeholder input, evident in Yang and Stansfield [18], are affirmed as a consideration needed to capture user needs and overall system effectiveness. However, as observed by Sinha et al. [16] and Feng et al. [8], neglecting this input may result in mismatches with user expectations. The analysis emphasizes the importance of stakeholder input into the design, response precision, and usability when developing chatbot and Q&A systems for education purposes. The analysis also highlights the reliance on overly complex systems, which are not easy to implement, update, and keep relevant in light of novel technologies and cannot be utilized by nontechnical audiences. The existing solutions also lack sufficient relevance for supporting module teaching, confirming a need for a system to unlock the potential for enhancing e-learning experiences.

G. Technology Frameworks

Hugging-face, a prominent open-source platform and library, offers an assortment of state-of-the-art NLP models, including transformer-based architectures GPT-2 and bidirectional encoder representations from transformers (BERTs).

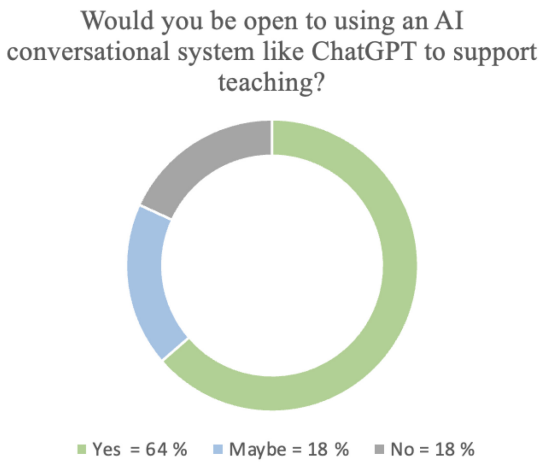

Fig. 1. Survey results for stakeholder openness to using GPT.

Renowned for enabling fine-tuning of these models, Hugging-face empowers developers to customize them for specific tasks, enhancing adaptability. However, OpenAI's GPT-3 takes precedence in this context due to its advanced NLP capabilities, which is informed by a vast and diverse training dataset. This equips GPT-3 to address a wide range of inquiries, including module-specific questions, while generating coherent and contextually relevant responses. OpenAI's user-friendly API streamlines GPT-3 integration, simplifying development and deployment processes. Thus, GPT-3's capabilities and ease of implementation, as well as the ability to contribute to the demystification of ChatGPT, position it as the preferred choice for the proposed system.

H. User Study Survey

A user survey (Figs. 1 and 2) was conducted among 11 members of academic stakeholders, including academics and teaching services, from the School of Electronic Engineering and Computer Science at QMUL, was conducted to gather insights and feedback on current pain-points for those responsible for education delivery. A detailed breakdown of the survey is outlined in Appendixes B and C in the supplementary material.

I. Technology Frameworks

The survey revealed that 82% of all participants across user segments were open to an implementation of a GPT-powered system. The need for task automation and routine support for academics is substantiated, with 83% of academics indicating daily use of QMPlus to support education delivery (which is greater than academic support staff averages of 10+ hours a week), highlighting the need for a system which can support "General Q&A" and "FAQs to stop general Q&A" is of utmost importance; further reflected in 60% of all participants agreed finding the time to respond to student queries a key challenge. 30% of participants perceive student engagement with forums and content as a key challenge, which corresponds with the requirement for real-time personalized responses, emphasising the importance of enhancing the learning experience. There was a clear preference raised for source attribution

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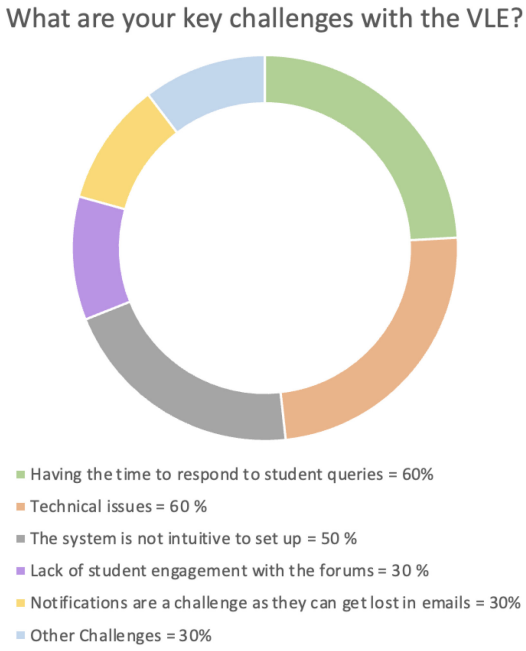

Fig. 2. Survey results for stakeholder insights into existing system limitations.

from multiple respondents and worries concerning incorrect or "hallucinating" responses, aligning with the requirement for transparency in the literature to ensure credibility and trust in the responses. In terms of demand for usability, 50% of all participants believe the system is not intuitive to use and resonates with the requirement for a user-friendly interface, ensuring ease of interaction for both students and staff. The technical challenges highlighted by 60% of all participants emphasize the urgency of efficient and dependable technological solutions, which came as a surprise as this was not a common theme in the literature, affirming the initial assumption about applications of novel technologies that are often developer-centric as opposed to user-centric. This highlights the need for rigorous testing and addressing reliability concerns to prevent potential setbacks before implementation, which was a concern of a teaching services respondent who does not want a new system to introduce "another point of potential failure." There are significant opportunities for potential future enhancements to the prototype that could address these existing pain points. For example, incorporating support for more complex question types based on technical problem-solving, as suggested by an academic respondent, could greatly benefit students. Additionally, there is potential to integrate the system into assessment use cases for technical question solving, further enhancing its value and impact. The intersection between the survey findings and the specified requirements reinforces the critical role of the proposed system in addressing the identified challenges and enhancing overall academic and student satisfaction.

III. System Design Methodology

Based on the findings in the literature, corroborated with user surveys, are the following requirements and user stories.

A. Functional Requirements

1) Q&A Functionality for Academic Support:

User Story: As an educator, I want routine tasks like addressing FAQs and aiding in module material understanding to be automated so that I can allocate more time to teaching students, research and preparing learning materials.

User Story: As a student, I want to be able to ask questions and receive relevant immediate responses to my queries that are tailored to my specific needs, ensuring a seamless and engaging learning experience.

2) Source Attribution for Credibility:

User Story: As a student, I appreciate transparency in learning materials. I want the system to provide clear source attribution to the information it presents, ensuring that I can trust the authenticity of the content.

3) Adaptability for Cross-Modules and Departments:

User Story: As an educator, I want the system to be adaptable to different modules and departments so that I can customize its responses and functionalities according to the specific educational needs and requirements of each module.

B. Nonfunctional Requirements

1) User Satisfaction:

User Story: As a student, I expect the system to respond promptly to my queries, providing personalized answers that align with my learning goals, ultimately contributing to my satisfaction with the educational experience.

User Story: As an educator, I want the proposed system to assist me in managing routine tasks efficiently so that I can focus on creating a rich academic environment, leading to enhanced satisfaction among my colleagues and students.

2) Stakeholder Confidence:

User Story: As academic support staff, I want to ensure that the integration of the technology is carried out with educator oversight, guaranteeing the system's reliability, accuracy, and alignment with educational objectives, thereby building stakeholder confidence.

3) Usability:

User Story: As a user of the tool, I want to interact with a user-friendly interface, ensuring ease of use and a positive user experience for students and staff.

C. SWOT Analysis of Potential Approaches

Appendix D in the supplementary material presents a strengths and weakness analysis comparing the fine-tuning and embedding approaches. In line with the findings of this analysis, the implementation of the proposed approach suits the embedding approach, specifically utilizing the pretrained language model GPT-3 and OpenAI's semantic vector text embedding features. This decision is supported by the benefits highlighted in the analysis, including that in adopting pre-trained models, the chatbot gains access to existing knowledge, cost-effectiveness, and versatility across various domains, as well as the ability to provide a broader language understanding beyond module materials. Moreover, the embedding approach facilitates rapid prototyping and development, aligning with

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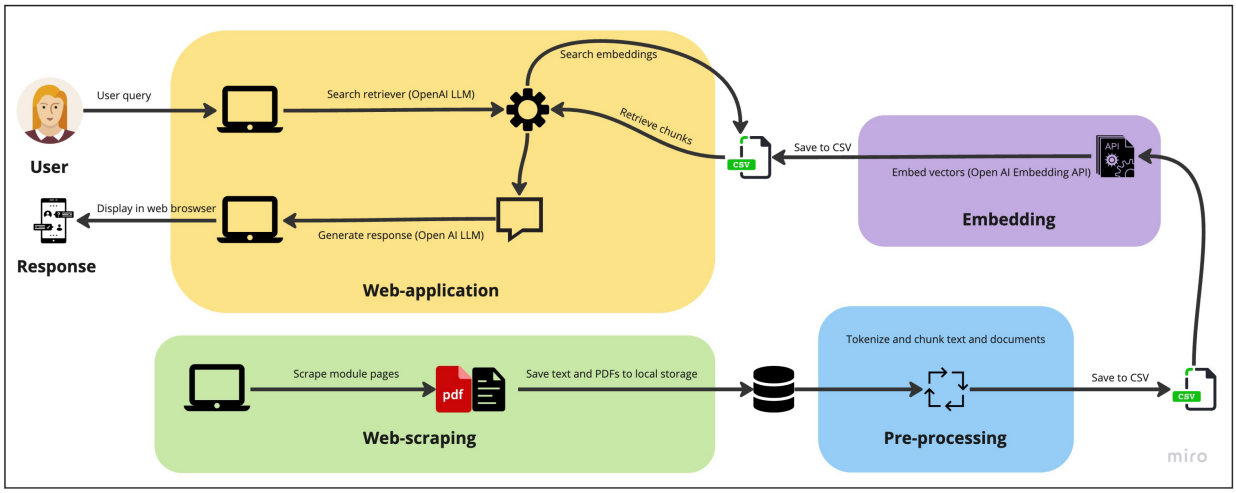

Fig. 3. System architecture of the prototype Q&A bot.

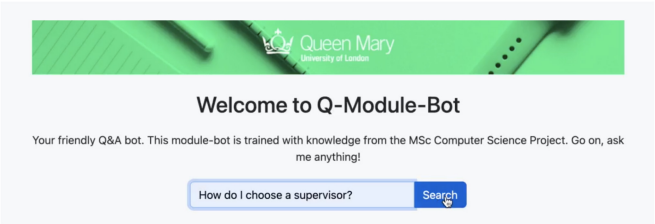

Fig. 4. Prototype search screen.

the implementation's goal of promoting awareness and discussing recommendations for conversational AI in education.

D. System Implementation

"Q-Module-bot" is designed to handle two main functions: 1) searching for relevant text chunks in the knowledge base and 2) generating responses using GPT-3 Turbo. The proposed system as shown in Fig. 3 has been developed in Python, which incorporates a data pipeline executed within a Jupyter Notebook environment, facilitating sourcing module content via Web-scraping, preprocessing, chunking, tokenization, and embedding utilizing OpenAI's Embedding API to generate dense vector representations of the data and store in a CSV file. This component ensures adaptability across modules by enabling easy integration of new knowledge through changes made exclusively to the embedding CSV file. A simple Web application, developed with Flask serves as the interface for user interaction and integrates with OpenAI's API, including the Q&A component powered by GPT-3.5 Turbo. This component handles user queries, generates responses, and facilitates natural language interaction.

E. Methodology

Fig. 4 shows the prototype search screen. The following are the main components.

1) *Web-Scraping:* This has been done by utilizing Beautiful Soup and Selenium, Python libraries used for Web scraping, which allows the system to extract module data from QMUL's VLE.

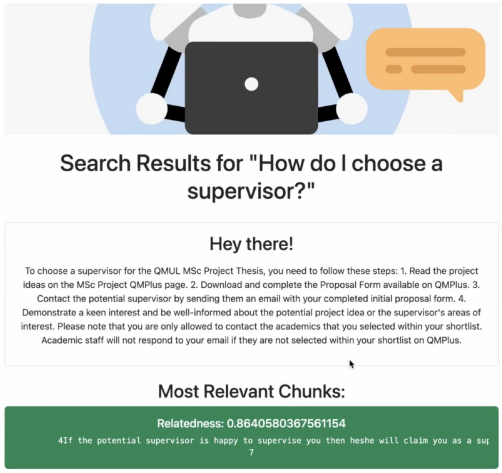

Fig. 5. Prototype Q&A bot response screen.

2) *Preprocessing:* Pandas and Matplotlib preprocess the data, removing noise, tokenizing, and formatting it into a CSV file.
3) *Embedding:* OpenAI's embeddings API transforms the tokenized data into dense vector representations, which are then stored in a local CSV file for use within the Web application.
4) *Application:* Flask's Web framework serves as the main interface of the application and loads data from a CSV file containing embeddings generated from text chunks. These embeddings are then preprocessed and converted from string representations to NumPy arrays, ensuring efficient computation. To search for relevant text chunks, the application implements a sophisticated ranking system based on the relatedness of the query to the content in the knowledge base; this process involves utilizing spatial distance computations to determine the similarity between embeddings. The application formulates queries for GPT-3.5 Turbo to generate responses, incorporating relevant source texts from the knowledge base (Fig. 5). When a user submits a query, the Web application constructs a tailored message for the GPT

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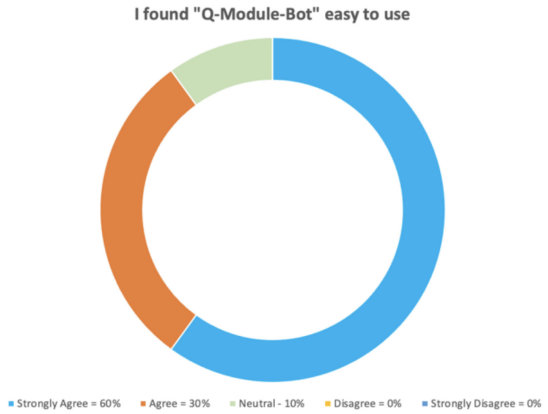

Fig. 6. Service evaluation response—Usability.

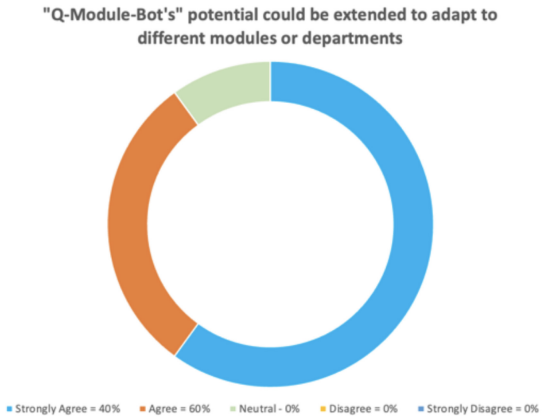

Fig. 7. Service evaluation response—Adaptability.

model, merging relevant source texts with the user's input. This ensures the GPT model has contextually relevant information for more informative responses. The Web application then displays these responses alongside relevant text chunks with relatedness scores of 80% and above displayed in green and below in red, giving users a comprehensive view of the retrieved information.

5) *Inquiries:* Q-Module-Bot has the capability to handle a wide range of queries, spanning from general information, such as assessment criteria and due dates, to more complex technical questions. However, the bot's ability to provide accurate responses is contingent on the information available in its knowledge base.

IV. Service Evaluation

To validate the system against the requirements and analyze the effectiveness of the system, a service evaluation was conducted with five academic stakeholders from the initial stakeholder group and five students from the M.Sc. project module used to test the system, which was deployed on Heroku. The group was asked to test the system by asking questions of their choice about this module. The participants were asked to anonymously answer a short questionnaire with the option for plain text feedback. The survey was created to reflect the initial requirements, with each question validating an initial requirement. A detailed breakdown of the results is presented in Appendix E in the supplementary material. The results were overwhelmingly positive; as seen in Fig. 6, 90% of users either strongly agreed or agreed that "Q-Module-Bot" was easy to use, 100% of users either strongly agreed or agreed that the text chunks and relatedness scores increased their confidence in the system, as reflected in Fig. 7, 100% of users either strongly agreed or agreed that the system could be extended to other modules and 100% of users either strongly agreed or agreed that the responses were relevant and helpful.

V. System Limitations

Noteworthy feedback emerged from the service evaluation that will contribute to future enhancements. In terms of user feedback, evaluators noted that the "relatedness score" was too lengthy and suggested minor user interface adjustments, such as optimization of font sizes for responses and configuration of the bot, to handle direction to a relevant human, e.g., when responses cannot be found.

The feedback also identified limitations in the evaluation process, as it focused on service assessment without a direct comparison to conventional search methods. The feedback also highlighted limitations in the evaluation process, which focused solely on assessing the application without directly comparing it to conventional search methods commonly used in teaching. This omission hindered a comprehensive understanding of the application's potential effectiveness. Additionally, as the current iteration of the application serves primarily as a proof of concept, technically, it is clear that the system lacks the robustness needed for widespread deployment. One notable area for improvement, as revealed in the evaluation, is the application's ability to handle complex queries effectively. While it can generate responses based on predefined prompts and source texts, its capability to understand nuanced queries, especially in technical contexts, is limited. Moreover, reliance on predefined source texts may restrict its ability to address queries and provide context beyond provided materials, though thorough prompt testing would be essential before integrating the application into academic environments to ensure appropriate responses and mitigate potential misuse. Furthermore, the application's lack of scalability due to local operation without robust database storage poses a significant obstacle, necessitating the implementation of scalable storage solutions for efficient data management and analysis. Additionally, updates to the user interface and full integration within the VLE system would enhance accessibility and user experience, further optimizing its effectiveness in educational settings. Q-Module-Bot, like other AI tools, may be affected by ethical issues and biases originating from the quality of input data, as the training data used for the module may be subjective. This is something that will be explored extensively in future iterations of this tool.

VI. Conclusion

To assess the achievement of the implementation's objectives, it is crucial to revisit the initial goals: enhancing student satisfaction, streamlining administrative tasks, and fostering the utilization of advanced AI in education. The feedback garnered from the service evaluation indicates a predominantly

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positive perception of "Q-Module-Bot." The fact that 80% of users either strongly agree or agree that they would use the bot again underscores a high level of student and academic content, reflecting their confidence in the bot's functionality and utility. However, the presence of neutral responses suggests an opportunity for further refinement. Overall, this feedback points to a favorable reception of "Q-Module-Bot" among the users who have engaged with it, which attests to its success in terms of promoting AI in education and increasing satisfaction levels.

In terms of alleviating administrative workload, while the evaluation encompassed a mixed user group of educators and students, the open-ended feedback suggests that enhancing user interface elements and ensuring the bot's responses are appropriately directed to humans would aid in reducing administrative burden. The feedback also suggested the need for an extended assessment period aligned with use in live module teaching. Thus, considering the overall positive reception, the potential for administrative relief, and acknowledging the limitations, the groundwork laid by this article encourages further exploration and development for a more robust and comprehensive implementation of "Q-Module-Bot."

There are also other benefits to the system, as outlined in the background research, which support fostering the utilization of advanced AI in education. This includes stakeholder engagement with the implementation, ensuring the prototype system aligns with the educational objectives and key stakeholders who will be responsible for its use and maintenance are involved throughout the process. Finally, the step-by-step guide in Appendix F in the supplementary material could be tested by academics to understand how extendable the application is and define specific improvements for use across modules.

Over the course of this work's timeline, the Generative AI landscape has undergone substantial transformations. It originated with limited information and API documentation, which meant time spent conducting technical research prior to programming the system. More recently, there has been a notable emergence of startups employing analogous Q&A systems across diverse sectors, including fields like HR, and plenty more supporting information and development documentation available. The captivating find of an active Facebook group composed of thousands of academics across the globe delving into the potentials of ChatGPT's applications has been particularly illuminating to follow throughout the work, reflecting the real interest and application of this work. This ever-evolving and dynamic milieu underscores the perpetual flux of the field, signifying a continual process of innovation and adjustment. The aim of this contribution is to serve as one of the foundational use cases in education with far-reaching adoption possibilities for audiences with varying levels of technical proficiency. The proposed approach places emphasis on both implementation practicality and ease, laying the groundwork for a broader exploration of innovative applications in education. Furthermore, the system maintains alignment with current teaching methods and documents available without requiring changes to existing teaching models.

Q-Module-Bot has the potential to bridge the gap between module content and student comprehension by providing personalized and automated support. The use of this tool can extend to various engineering courses with substantial content loads and to modules where students frequently require clarification on complex topics.

Future Work

There are ample opportunities for refining and enhancing the functionality of "Q-Module-Bot" based on user feedback. The initial user studies shed light on technical glitches within existing systems, highlighted by 60% of academics, reinforcing the need for robust and dependable technological solutions. This underscores the importance of comprehensive testing to ensure reliability and mitigate any potential points of failure before considering implementation in university systems. Additionally, several avenues for future enhancements are evident, addressing existing shortcomings and catering to specific needs, including the addition of training materials to the knowledge base. One promising enhancement involves enabling the system to handle more intricate question types that revolve around technical problem-solving, integrating the system into assessment scenarios holds significant potential, extending its utility beyond its current scope [36]. Furthermore, work could be done to only include text chunks in the response formulation if over a specific "relatedness" threshold (e.g., if over 80% confidence).

A. Advanced Generative AI Models

The long term objective is that Q-Module-Bot will be updated regularly with the latest Generative AI models to improve the performance in terms of accuracy of the responses generated, which will be validated using precision and recall [38].

B. Experiments With Larger Sample Size

The true potential of the system will become clear when it is used as an integral part of university module teaching throughout an academic term. Beyond its immediate benefits as a Q&A system, an exciting prospect emerges—leveraging the wealth of data it generates for deeper insights. By analyzing this data, we can not only identify areas of content that students find most challenging but also gain a nuanced understanding of their engagement levels [26]. This data-driven approach serves as a valuable metric for the overall health of modules and ultimately academic institutions, offering actionable insights to inform decision-making and targeted teaching service interventions. These insights could empower educators and administrators to proactively address issues, refine teaching strategies, and optimize the learning experience. The journey ahead holds transformative possibilities for educational technology. By embracing analytics and using the system as a catalyst for meaningful change, we pave the way for a future where education is continually enhanced through innovative solutions and evidence-based approaches.

In terms of the service evaluation, a limitation we encountered was related to the small sample size of the stakeholders involved. To address this issue, we intend to employ interpretative research inquiry in the form of semi-structured interviews with the stakeholders [37]. By doing so, we aim to explore

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their detailed feedback on the affordances, which will allow us to gain a deeper understanding of their experience and observations. Furthermore, this approach will enable us to identify areas that require improvement, particularly with regard to Q-Module-Bot's architecture, data handling, and integration processes.

In addition to semi-structured interviews, our objective is to conduct an extensive experiments of Q-Module-Bot using a larger sample of university students. The purpose of this experiments will be to evaluate the effectiveness and usability of Q-Module-Bot in answering questions related to the module content compared to traditional methods, such as asking module teaching staff or searching manually through the VLE. For this evaluation, we aim to recruit around 100 students enrolled on a module where Q-Module-bot will be trained. The students will be randomly assigned into a control and experimental group. For the control group, the students will use traditional methods (such as VLE or module teaching team) to find answers to their questions. While the experimental group, will use the Q-Module-Bot to find their answers. The experiment will be conducted over a 12 week period (1 term), which will be followed a usability survey that will need to be completed by both groups to evaluate the perceived ease of use, satisfaction and effectiveness.

C. Data Privacy and Ethics

We anticipate that this research and future experiments will enhance the overall performance of the service and improve the user experience. On the whole, the current limitations of the "Q-Module-Bot" have the potential to be addressed through fine-tuning without necessitating extensive retraining. As we progress, it is imperative to consider ethical concerns that arise regarding data privacy, such as issues surrounding access, to data and its potential misuse, which must be communicated transparently to end-users. Upon overcoming the system's current limitations and ethical concerns, there is an opportunity to extend its scope by implementing "Q-Module-Bot" within courses and modules across various disciplines in the institution. This would serve as a catalyst for conducting longitudinal studies to evaluate the impact of this tool on educational outcomes over a more extended period. Such an investigation will contribute to a more nuanced understanding of the long-term benefits of incorporating AI-powered assistants in education.

Software Availability

We released Q-Module-Bot as an open source software and the implementation code with experiment scripts and results can be found at the GitHub repository: https://github.com/iamssgill/Q-Module-Bot

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